



RootED

Keeping first-generation learners in the classroom.

A holistic learn-and-stay ecosystem that tackles the real reasons students drop out — language, money, learning gaps, and what's happening at home.

HACKATHON IDEA SUBMISSION · TEAM ROOTED



Why first-generation learners drop out

THE PROBLEM WE'RE SOLVING

They're often the first in their family to study this far. When they hit a wall, there's no one at home who has been there before — so small problems quietly snowball into dropping out.



Money pressure

Family needs them to earn now, not study for a far-off payoff.



Language barrier

Best lectures are in English; concepts get lost in translation.



Compounding gaps

One missed basic makes every later topic feel impossible.



Unstable home

No quiet space, no guidance, sometimes a hostile environment.



No support net

No mentor to ask, no idea which scholarships even exist.



Low family buy-in

Parents underestimate the long-term value of finishing.



Our idea: meet every barrier with a response

THE BIG PICTURE

Most ed-tech just adds more content. RootED is different — for each reason a student leaves, we built a specific pillar to keep them in.

Can't afford to keep studying



Learn & Earn — get paid to translate lectures

Lessons aren't in my language



Native-language transcripts of top open lectures

I'm lost and falling behind



Gamified gap-finder + a quick personal roadmap

Nobody tells me what's out there



Reels feed: bite-size content, quizzes, scholarships

My family wants me to quit



NGO-led family counseling on long-term payoff

Home is not a place I can learn



Affordable hostel / housing access

I have a problem and no one to ask



RAG chatbot that finds the exact help + links



Six pillars, one platform

WHAT WE'RE BUILDING



1. Learn & Earn

Verified students translate open lectures into native languages — and earn micro-income for every vetted transcript.



2. Gap-Finder

A short, gamified diagnostic finds the root gap behind your confusion and builds a quick roadmap to close it.



3. Reels Feed

Swipeable short-form: chapter recaps, quick quizzes and scholarship alerts you can share with friends.



4. Family Bridge

NGOs counsel families on the long-term value of education and how to support their learner.



5. Safe Housing

Access to affordable hostels so a tough home situation doesn't end an education.



6. Help Chatbot

Ask anything in your language; a RAG assistant retrieves the exact resource, scheme or link — or routes you to a human.



Built for VIDYA — all five sub-points

THEME ALIGNMENT

We're targeting the VIDYA theme. Here's how the platform answers every sub-point it asks for — skilling and returnees included.



First-generation learner support

The whole platform is designed around them — every pillar removes one barrier they face.



Learning in mother tongue

Native-language lecture transcripts via AI4Bharat ASR + IndicTrans2.



School dropout prevention

Gap-Finder, Family Bridge and safe housing work together to keep students enrolled.



Vocational skill documentation

Learn & Earn doubles as a verifiable 'skills passport'; reels carry vocational micro-content too.



Bridge learning for returnees

Gap-Finder builds a catch-up roadmap so dropouts can rejoin at the right level.



How it all fits together

SYSTEM FRAMEWORK

WHO USES IT

Students

Families

Student-translators

NGOs & mentors



RootED app · mobile-first · multilingual · works on low bandwidth & offline



Content & Translation

transcribe · translate · vet



Learning Diagnostics

gap-finder · roadmaps



Support & RAG

chatbot · grievances



Community & Incentives

payouts · NGOs · reels

FOUNDATIONS AI4Bharat ASR + IndicTrans2 · Vector DB (RAG) · Concept graph · UPI payouts · Cloud + CDN



Pipeline 1 — Learn & Earn

FEATURE 1 · TRANSCRIPTION

Turn the world's best free lectures into native-language transcripts — and pay the students who make them.



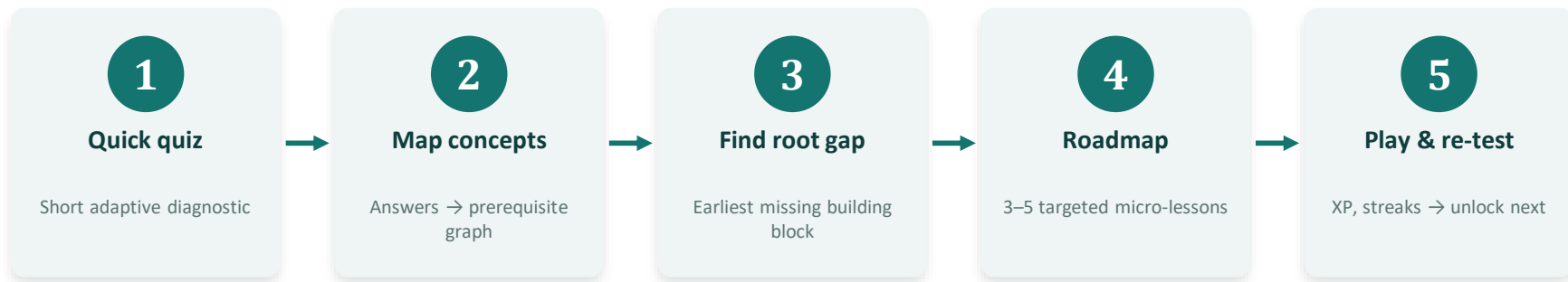
Why it works: the student earns while learning, instead of choosing between the two. Every vetted transcript is also documented, credentialed work — a verifiable 'skills passport' they can show employers. AI drafts; humans guarantee the quality.



Pipeline 2 — Gamified Gap-Finder

FEATURE 2 · DIAGNOSTICS

Students rarely fail because of today's topic — they fail because of a basic they missed long ago. We find that root gap and close it.



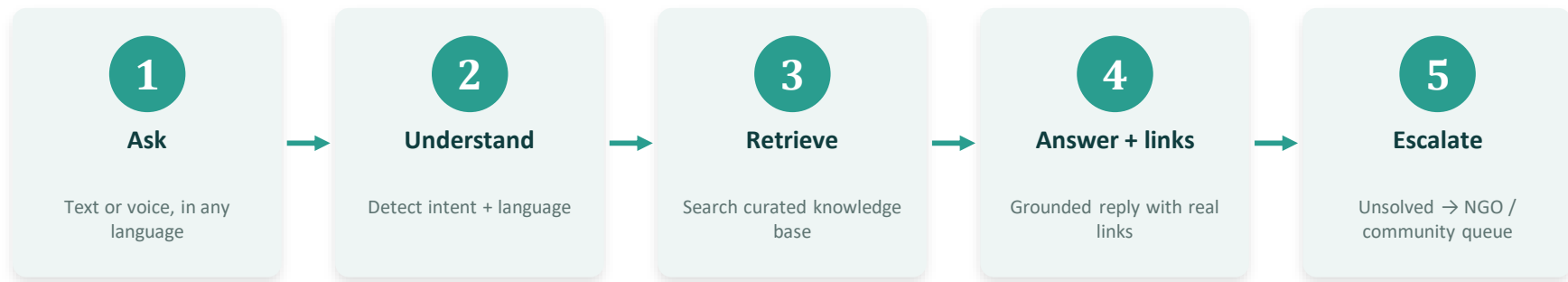
Why it works: instead of generic re-teaching, we pinpoint the one early concept causing the snowball and make fixing it feel like a game. The same engine builds a catch-up 'bridge' roadmap for returnees — students who dropped out and want back in at the right level.



Pipeline 3 — Help & Grievance Chatbot

FEATURE 6 · RAG ASSISTANT

A student with a problem and no one to ask is a student about to leave. This assistant always has an answer — or a human who does.



Knowledge base: scholarships, government schemes, study resources, counseling and helpline contacts. RAG keeps answers accurate and link-backed — no made-up advice — and unresolved needs get posted for a real person to help.



Beyond the app: people and place

PILLARS 3 · 4 · 5

Technology keeps a student engaged. People and a safe place keep them enrolled. These three pillars run on partnerships, not just code.



Reels feed

Swipeable chapter recaps, 30-second quizzes and scholarship alerts. Shareable, so good content spreads through friend groups the way social media already does.



Family Bridge

NGO partners run short counseling sessions with families — explaining the long-term payoff of finishing, and how to back their learner instead of pulling them out.



Safe Housing

Tie-ups for affordable hostel and housing access, so a tough or unsafe home doesn't decide a student's future. Learning should not depend on luck of birth.



How we'll build it

TECH STACK



Front-end

Flutter / React Native

Mobile-first, offline cache

Low-data 'lite' mode



AI & language

AI4Bharat ASR (IndicWhisper)

IndicTrans2 translation

LLM + embeddings for RAG



Data & logic

FastAPI + PostgreSQL

Vector DB (Chroma / FAISS)

Neo4j concept graph



Trust & money

Phone-OTP auth

Light KYC for payouts

UPI micro-payments



What we'll build this weekend

SCOPE — STAYING REALISTIC

We can't ship all six pillars in a hackathon — so we'll build a working core that proves the idea, and pitch the rest as the roadmap.

✓ BUILD & DEMO NOW

- Gap-Finder: diagnostic → root gap → roadmap
- RAG chatbot seeded with real scholarships & schemes
- Transcription demo: 1 lecture → ASR → translate → review UI → mock payout
- Reels feed prototype with sample quiz cards

→ PITCH AS ROADMAP

- Real UPI payouts & translator verification at scale
- NGO partnerships for family counseling
- Affordable housing / hostel tie-ups
- Scholarship & government-scheme integrations



From hackathon to real impact

NOW

Hackathon MVP

Gap-Finder, RAG chatbot, transcription demo & reels prototype.

3–9 mo

Grow

More languages & lectures, onboard NGOs for counseling, plug in scholarship data.

0–3 mo

Pilot

Run with 1–2 colleges + an NGO. Seed content in 2 languages, switch on real payouts.

9 mo +

Scale

Housing tie-ups, CSR & government funding, multi-state rollout.



Why RootED works

RootED · Team RootED

- ✓ Attacks root causes — money, language, gaps and home — not just content.
- ✓ Native-language first, mobile-first, built for low bandwidth.
- ✓ Earn-while-you-learn breaks the 'drop out to earn' trap.
- ✓ AI does the heavy lifting; people and NGOs build the trust.

Retention fewer dropouts

Mastery pre/post gains

₹ earned by students

Resolved grievances